Mathematical Logic

- **Exercise 1**
  Translate to symbolic logic the following statement defining propositions and using appropriated connectives: $\sin\left(\frac{\pi}{4}\right) > 0$ and $\sin\left(\frac{\pi}{4}\right) < 1$,
    - **Solution 1.1**
      Let:
      $p : \sin\left(\frac{\pi}{4}\right) > 0$
      $q : \sin\left(\frac{\pi}{4}\right) < 1$

      The symbolic logic is:
      $$p \wedge q$$

- **Exercise 2**
  Translate to symbolic logic the following statement defining propositions and using appropriated connectives: 9 is a perfect square or 8 is a perfect square.
    - **Solution 2.1**
      Let:
      $p : 9$ is a perfect square.
      $q : 8$ is a perfect square.

      The symbolic logic is:
      $$p \vee q$$

- **Exercise 3**
  Translate to symbolic logic the following statement defining propositions and using appropriated connectives: The sum of a and b is greater than zero or zero and smaller than the product of them.
    - **Solution 3.1**
      Let:
      p : The sum of a and b is greater than zero ($a + b > 0$).
      q : Zero is smaller than the product of them ($0 < ab$).

      The symbolic logic is:
      $$p \vee q$$

- **Exercise 4**
  Translate to symbolic logic the following statement defining propositions and using appropriated connectives: If P is a parabola, then the equation of P is quadratic.
    - **Solution 4.1**
      Let:
      $p : P$ is a parabola.
      q : The equation of P is quadratic.

      The symbolic logic is:
      $$p \implies q$$

- **Exercise 5**

Translate to symbolic logic the following statement defining propositions and using appropriated connectives: $x-2$ is greater than zero and $x+2$ is smaller than zero or $x-2$ is smaller than zero and $x+2$ is greater than zero.

- **Solution 5.1**
  Let:
  $p: x-2$ is greater than zero ($x-2>0$).
  $q: x+2$ is smaller than zero ($x+2<0$).
  $r: x-2$ is smaller than zero ($x-2<0$).
  $s: x+2$ is greater than zero ($x+2>0$).

  The symbolic logic is:

$$(p \wedge q) \vee (r \wedge s)$$

- **Exercise 6**
  Translate to symbolic logic the following statement defining propositions and using appropriated connectives: $(x+2)(x-2)$ is smaller than zero.
  - **Solution 6.1**
    Let:
    $p: (x+2)(x-2)$ is smaller than zero.

    The symbolic logic is:

$$p$$

- **Exercise 7**
  Translate to symbolic logic the following statement defining propositions and using appropriated connectives: It is not true that $x-3$ is greater than zero or smaller than 1.
  - **Solution 7.1**
    Let:
    $p: x-3$ is greater than zero ($x-3>0$).
    $q: x-3$ is smaller than 1 ($x-3<1$).

    The symbolic logic is:

$$\neg(p \vee q)$$

- **Exercise 8**
  Translate to symbolic logic the following statement defining propositions and using appropriated connectives: If f is not a constant, then f' is not zero.
  - **Solution 8.1**
    Let:
    $p: f$ is a constant.
    $q: f'$ is zero.

    The symbolic logic is:

$$\neg p \implies \neg q$$

- **Exercise 9**
  Translate to symbolic logic the following statement defining propositions and using appropriated connectives:

If $\sin x$ is 0 and x is greater than, or equal to 0 and smaller than or equal to 2π, then $x = 0$ or $x = \pi$.

- **Solution 9.1**
  Let:
  $p : \sin x$ is 0.
  $q : x$ is greater than, or equal to 0 and smaller than or equal to 2π ($0 \leq x \leq 2\pi$).
  $r : x = 0$.
  $s : x = \pi$.

  The symbolic logic is:

  $$(p \wedge q) \implies (r \vee s)$$

- **Exercise 10**
  Translate to symbolic logic the following statement defining propositions and using appropriated connectives:
  If $\cos x \cdot \sin x$ is 0 and x is greater than 0 and smaller than, or equal to π, then $x = \frac{\pi}{2}$.

  - **Solution 10.1**
    Let:
    $p : \cos x \cdot \sin x$ is 0.
    $q : x$ is greater than 0 and smaller than, or equal to π ($0 < x \leq \pi$).
    $r : x = \frac{\pi}{2}$.

    The symbolic logic is:

    $$(p \wedge q) \implies r$$

- **Exercise 11**
  Let p stand for the proposition "$x^2 + 2x - 3$ is 0" and q for "$x^2 - x - 2$ is 0". Express the following as natural sentences in English:

  $$\begin{gathered} \neg p \\ p \wedge q \\ p \vee q \end{gathered}$$

  - **Solution 11.1**

    $$\begin{gathered} x^2 + 2x - 3 \text{ is not } 0. \\ x^2 + 2x - 3 \text{ is } 0 \text{ and } x^2 - x - 2 \text{ is } 0. \\ x^2 + 2x - 3 \text{ is } 0 \text{ or } x^2 - x - 2 \text{ is } 0. \end{gathered}$$

High school mathematical concepts with quantified definitions

- **Exercise 14**
  Perfect square
  An integer k is a perfect square if there exists an integer n such that $k = n^2$.

  - **Solution 14.1**
    The statement can be expressed as:

    $$\exists n \in \mathbb{Z},\ k = n^2$$

- **Exercise 15**
  Odd number

An integer m is odd if there exists an integer t such that $m = 2t + 1$.

- **Solution 15.1**
  The statement can be expressed as:

$$(\exists t \in \mathbb{Z})\, m = 2t + 1$$

- **Exercise 16**
  Multiple
  An integer n is a multiple of the integer d if there exists an integer k such that $n = dk$.
  - **Solution 16.1**
    The statement can be expressed as:

$$(\exists k \in \mathbb{Z}),\ n = dk$$

- **Exercise 17**
  Rational number
  A real number x is rational if there exist integers a, b with $b \neq 0$ such that $x = \frac{a}{b}$.
  - **Solution 17.1**
    The statement can be expressed as:

$$(\exists a \in \mathbb{Z})(\exists b \in \mathbb{Z})\, b \neq 0,\ x = \frac{a}{b}$$

- **Exercise 18**
  Irrational number
  A real number x is irrational if for all integers a, b with $b \neq 0$, we have $x \neq \frac{a}{b}$.
  - **Solution 18.1**
    The statement can be expressed as:

$$(\forall a, b \in \mathbb{Z})\, b \neq 0 \implies x \neq \frac{a}{b}$$

- **Exercise 19**
  Root (zero) of a function
  A real function $f : \mathbb{R} \to \mathbb{R}$ has a root if there exists a number $c \in \mathbb{R}$, such that $f(c) = 0$.
  - **Solution 19.1**
    The statement can be expressed as:

$$(\exists c \in \mathbb{R})\, f(c) = 0$$

- **Exercise 20**
  Function bounded above
  A real function $f : \mathbb{R} \to \mathbb{R}$ is bounded above if there exists a real number $M > 0$, such that for all $x \in \mathbb{R}$, $f(x) \leqslant M$.
  - **Solution 20.1**
    The statement can be expressed as:

$$(\exists M \in \mathbb{R}^+)(\forall x \in \mathbb{R})\, f(x) \leqslant M$$

- **Exercise 21**
  Function bounded below
  A real function $f:\mathbb{R}\to\mathbb{R}$ is bounded below if there exists a real number m such that for all $x\in\mathbb{R}$, $f(x)\geqslant m$.
  - **Solution 21.1**
    The statement can be expressed as:

$$(\exists m\in\mathbb{R})(\forall x\in\mathbb{R})\, f(x)\geqslant m$$

- **Exercise 22**
  Strictly increasing function
  A real function $f:\mathbb{R}\to\mathbb{R}$ is strictly increasing if for all $x_1, x_2\in\mathbb{R}$, $x_1 < x_2$, we have $f(x_1) < f(x_2)$.
  - **Solution 22.1**
    The statement can be expressed as:

$$(\forall x_1, x_2),\ x_1 < x_2 \implies f(x_1) < f(x_2)$$

- **Exercise 23**
  Equal functions
  A real function $f:\mathbb{R}\to\mathbb{R}$ equals a real function $g:\mathbb{R}\to\mathbb{R}$ if for every $x\in\mathbb{R}$, $f(x)=g(x)$.
  - **Solution 23.1**
    The statement can be expressed as:

$$(\forall x\in\mathbb{R})\, f(x)=g(x)$$

- **Exercise 24**
  Different functions
  A real function $f:\mathbb{R}\to\mathbb{R}$ is different from a real function $g:\mathbb{R}\to\mathbb{R}$ if there exists $x\in\mathbb{R}$, $f(x)\neq g(x)$.
  - **Solution 24.1**
    The statement can be expressed as:

$$(\exists x\in\mathbb{R})\, f(x)\neq g(x)$$

- **Exercise 25**
  Congruent triangles
  Two triangles are congruent if there exists a rigid transformation T mapping one triangle onto the other.
  - **Solution 25.1**
    The statement can be expressed as:

$$(\exists \text{rigid motion } T)\, T(\Delta_1)=\Delta_2$$

- **Exercise 26**
  A very well-known Theorem of High-School
  Let $f(x)=ax^2+bx+c$, $a, b, c\in\mathbb{R}$ be any quadric.

  (a) If $b^2-4ac<0$, then f doesn't have real roots.
  (b) If $b^2-4ac=0$, then f has exactly one real root.
  (c) If $b^2-4ac>0$, then f has exactly two real roots.
  - **Solution 26.1**

For statement (a), it can be expressed as:

$$(\forall x \in \mathbb{R})\, f(x) \neq 0$$

- **Solution 26.2**
  For statement (b), using the unique existential quantifier ∃!:

  $$(\exists! x)\, f(x) = 0$$

  Alternatively, without using ∃!:

  $$(\exists x)(\forall y)\, (f(x) = 0 \land (y \neq x \implies f(y) \neq 0))$$

- **Solution 26.3**
  For statement (c), it can be expressed as:

  $$(\exists x)(\exists z)(\forall y)\, (x \neq z \land f(x) = 0 \land f(z) = 0 \land (y \neq x \land y \neq z \implies f(y) \neq 0))$$